ROHAN GARG

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PROFESSIONAL SUMMARY

Software & AI/ML Developer Agentic AI systems, full-stack apps, and scalable backend infrastructure. Skilled in modern frameworks, like langchain and langgraph, and making RAG pipelines. Seeking an SDE or AI/ML Engineering role to build softwares

PROJECTS

CivicEye**: (Python, PyTorch, YOLO11m, OpenCV, SAHI, Roboflow, Multi-threading)**
Real-Time Edge AI Surveillance System | Lead Systems & AI Architect

- Built a zero-latency Edge AI pipeline for real-time micro-litter detection on consumer CCTV hardware.
- Engineered a dual-threaded YOLO/SAHI Asynchronous Pipeline to bypass 2-FPS latency bottlenecks and maintain 30+ FPS.
- Resolved PyTorch FP16 memory overflows on mobile GPUs by enforcing FP32 precision, eliminating spatial hallucinations.
- Executed Hard Negative Mining to suppress AI false positives on complex background textures to near zero.
- Deployed inference entirely on local Edge nodes, eliminating cloud dependency and 1.5s API network lag.

AI Career Copilot**: (Python, FastAPI, React, PostgreSQL, LLMs, RAG, Embeddings)**
AI-powered career guidance platform that analyzes resumes, skills, projects, and professional profiles to generate personalized career roadmaps.

- Built a FastAPI-based resume processing pipeline with PDF text extraction and structured profile analysis.
- Implemented **RAG, embeddings, and LLMs** for semantic skill-gap analysis and personalized career recommendations.
- Designed a scalable **React + FastAPI + PostgreSQL** architecture for AI-driven career planning and recommendations.

EDUCATION

- ***Bachelor of Technology in Computer Science and Engineering (Bennett University)*** *2024 – 2028*
- ***Translam Academy International*** *2020 – 2024*

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C++, JavaScript, TypeScript, Node.js, ReactJS, FastAPI, PyTorch
- **Databases:** MongoDB, MySQL, PostgreSQL, Qdrant
- **DevOps:** Docker, CI/CD, Render, Supabase, Cloudflare
- **Version Control & Hosting:** Git, GitHub, GitHub Actions, Vercel
- **Artificial Intelligence:** Machine Learning, Deep Learning, Neural Networks, Image Processing, Large Language Models (LLMs), LangChain, LangGraph, RAG
- **Systems:** Monolithic, Microservices
- **CS Fundamentals:** DBMS, Operating Systems, Computer Networks, OOPs

EXPERIENCE AND CERTIFICATES

Bachelor of Technology in Computer Science and Engineering (Bennett University) *2024 – 2025*
Foresight X — AI-Powered Customer Retention Platform | Hackathon Project
*Developed an end-to-end customer churn prediction platform using **Python, FastAPI, React, PostgreSQL, Scikit-learn, XGBoost, SHAP, and MLflow**. Built leakage-aware customer snapshots, compared multiple ML models, generated explainable churn-risk scores, and prioritized customers using predicted business exposure to support data-driven retention decisions.*